

Statistics for Data Science - I
Sept 2025 Term

1. In a sports analytics company, the team is interested in predicting whether a basketball player will score above 20 points in a game using classification models. They collected data on various factors $\mathbf{x} = [x_1, x_2, x_3]$ from past games of different players. The factors include minutes played, shots attempted, and free throw percentage. The data and corresponding labels are shown in the table below:

$\mathbf{x}$	y
[1,5,0.8]	1
[0,3,0.9]	0
[1,2,0.7]	0
[1,8,0.6]	1
[0,7,0.5]	0

Handwritten notes next to the table:

- $\frac{1}{5}$ (circled)
- [1, 5, 0.8]
- [0, 3, 0.9]
- [1, 2, 0.7]
- [1, 8, 0.6]
- [0, 7, 0.5]

$$\text{loss} = \frac{1}{n} \sum_{i=1}^n I(f(x^i) \neq y^i)$$

Compute the **misclassification rate** if they use the following step function:

$$u(z) = \begin{cases} 1, & \text{if } z \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

and the linear combination is given by:

$$z = 0.6x_1 + 0.3x_2 + 0.1x_3$$

Here, x_1 represents whether the player is a starter (1) or not (0), x_2 is the number of shots attempted, and x_3 is the free throw percentage. The label y indicates whether the player scored more than 20 points (1) or not (0).

The misclassification rate is the proportion of labels that are incorrectly classified. For example, if 3 out of 5 players are mislabeled, then the misclassification rate is $\frac{3}{5}$. **Enter the answer correct to one decimal place.**

Answer : 0.2

$\mathbf{x}$	$z = 0.6x_1 + 0.3x_2 + 0.1x_3$	y	$u(z)$
[1, 5, 0.8]	2.18	1	0
[0, 3, 0.9]	0.99	0	0
[1, 2, 0.7]	1.27	0	0
[1, 8, 0.6]	3.06	1	1
[0, 7, 0.5]	2.15	0	0

$$Loss = \frac{1}{5} (1) = \frac{1}{5} = 0.2$$

2. Consider a dataset from an e-commerce company such as *Flipkart* that has details of customer behavior. Which of the following problems can be best approached using **clustering**?
- A. Predicting the exact number of sales for each customer segment.
 - B. Classifying customers into segments based on labeled customer data.
 -  C. Identifying inherent groupings in customer data without predefined labels.
 - D. Reducing the number of features in the customer dataset.

Answer : c

3. For what value of b is the following function continuous at every x ?

$$f(x) = \begin{cases} \frac{x-b}{b+1}, & x < 0 \\ x^2 + b, & x \geq 0 \end{cases}$$

A. 3

B. 2

✓✓✓ C. -2 ✓

D. 1

Answer : c

$$f(a) = \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) \quad \text{--- } (*)$$

$$\lim_{x \rightarrow 0^-} f(x) = \frac{0-b}{b+1} = \frac{-b}{b+1}$$

$$\lim_{x \rightarrow 0^+} f(x) = 0^2 + b = b$$

$$\Rightarrow \frac{-b}{b+1} = b$$

$$\Rightarrow -b = b(b+1)$$

$$\Rightarrow -b = b^2 + b$$

$$\Rightarrow b^2 + 2b = 0$$

$$\Rightarrow b(b+2) = 0$$

$$\Rightarrow b = 0 \text{ or } \boxed{b = -2}$$

4. The function

$$f(x) = \begin{cases} ax^2 + bx + 1, & x < 0, \\ cx + d, & x \geq 0, \end{cases}$$

is differentiable at $x = 0$. If a, b, c, d are constants, which of the following must be true?

A. $a = c$ and $b = 1$

B. $b = c$ and $d = 1$

C. $a = c$ and $d = 1$

D. $b = d = 1$

Answer : b

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$ continuity.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (ax^2 + bx + 1) = 1$$

$$\lim_{x \rightarrow 0^+} f(x) = d.$$

$$f(0) = d$$

$$\Rightarrow d = 1$$

$$\lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \left. \frac{d}{dx} (ax^2 + bx + 1) \right|_{x=0}$$

$$= 2ax + b \Big|_{x=0} = b$$

$$\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \left. \frac{d}{dx} (cx + d) \right|_{x=0} = c.$$

$$b = c$$

5. Consider the function $f(x, y) = \sin(xy)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ around the point $(\frac{\pi}{2}, 0)$. What is the value of $L(\frac{\pi}{2} + 0.1, 0.1)$? (Use the approximation $\pi = 3.14$)

A. 0.162

B. 0.257

C. 0.157

D. 1.57

✓✓✓ Answer : c ✓✓

$$L(x, y) = \underbrace{f(a, b)} + \underbrace{\frac{\partial f}{\partial x}(a, b)} \cdot (x - a) + \frac{\partial f}{\partial y}(a, b) \cdot (y - b)$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} [\sin(xy)] = \underbrace{y \cos(xy)}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} [\sin(xy)] = x \cos(xy)$$

$$f(x, y) = \sin(xy)$$

$$f\left(\frac{\pi}{2}, 0\right) = \sin\left(\frac{\pi}{2} \cdot 0\right) = \sin(0) = \underline{0}$$

$$\frac{\partial f}{\partial x}\left(\frac{\pi}{2}, 0\right) = 0 \cdot \cos(0) = 0$$

$$\frac{\partial f}{\partial y}\left(\frac{\pi}{2}, 0\right) = \frac{\pi}{2} \cdot \cos(0) = \frac{\pi}{2}$$

Now, substitute in eqⁿ ①,

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b) \cdot (x - a) + \frac{\partial f}{\partial y}(a, b) \cdot (y - b)$$

$$L(x, y) = \underbrace{f\left(\frac{\pi}{2}, 0\right)} + \frac{\partial f}{\partial x}\left(\frac{\pi}{2}, 0\right) \cdot \left(x - \frac{\pi}{2}\right) + \frac{\partial f}{\partial y}\left(\frac{\pi}{2}, 0\right) \cdot (y - 0)$$

$$= 0 + 0 \cdot \left(x - \frac{\pi}{2}\right) + \frac{\pi}{2} \cdot y = \frac{\pi y}{2}$$

$$L(x, y) = \frac{\pi}{2} (y)$$

$$L\left(\frac{\pi}{2} + 0.1, \underline{0.1}\right) = \frac{\pi}{2} \times (0.1) = 1.57 \times 0.1 = 0.157.$$

maximum directional derivative occurs in the direction of the gradient vector, normalized.

6. Consider the function $f(x, y) = x^2 + 3y^2$. In which direction does the maximum directional derivative occur at the point $(1, 2)$?

✓ A. $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$

✓ B. $\left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}}\right)$

✓✓✓ C. $\left(\frac{2}{\sqrt{5}}, \frac{24}{\sqrt{5}}\right)$

D. $\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

Answer : b

$$\nabla f(1, 2) = (2, 12)$$

$$\text{Magnitude of } (2, 12) = \sqrt{2^2 + 12^2} = 2\sqrt{37}$$

Unit vector is:

$$\frac{1}{2\sqrt{37}} (2, 12) = \left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}}\right)$$

$$x^T y = 0, \quad x^T z = 0$$

7. Which of the following options is/are true?

✓ A.

$x \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is orthogonal to the vectors $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

⊗

B. For a matrix A , vectors of the column space of A are perpendicular to the vectors of the null space of A .

✓ C.

Length of the vector

$$a = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \text{ is } 3.$$

✓✓ D.

$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ are orthogonal vectors.

Answer : a, d

$$x^T y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 = 0.$$

$$x^T z = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 0 + 0 + 0 = 0.$$

$$\text{length} = \|a\| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}$$

$$x^T y = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = 1 \cdot 1 + 2 \cdot 1 + 3 \cdot (-1) = 3 - 3 = 0.$$

(a)

$$\frac{a}{\|a\|}$$

$$\begin{pmatrix} 2 & 5 \\ 6 & 15 \end{pmatrix} R_2 \rightarrow R_2 - 3R_1 \quad \begin{pmatrix} 2 & 5 \\ 0 & 0 \end{pmatrix}$$

8.

Consider a matrix $A = \begin{pmatrix} 2 & 5 \\ 6 & 15 \end{pmatrix}$.

3 x row 1
rank(A) = 1

Which of the following options is/are true?

✓✓✓ A. Null space of A is span $\left\{ \begin{pmatrix} -5 \\ 2 \end{pmatrix} \right\}$ ✓ → Null space. $AX=0$.

✓✓✓ B. Column space of A is span $\left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\}$ ✓

✓✓ C. Left null space of A is span $\left\{ \begin{pmatrix} -3 \\ 1 \end{pmatrix} \right\}$ ✓

✗ D. Row space of A is span $\left\{ \begin{pmatrix} 2 \\ -5 \end{pmatrix} \right\}$ ✗

Answer : A, B, C

$AX=b$

$$\begin{pmatrix} 2 & 5 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

→ $2x_1 + 5x_2 = 0$

⇒ $x_1 = -\frac{5}{2}x_2$

if $x_2 = 2$,

$x_1 = -5$

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

2 $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

(III)

Solve $A^T y = 0$

$$\begin{pmatrix} 2 & 6 \\ 5 & 15 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 6 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$R_2 \rightarrow R_2 - 2.5R_1$

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⇒ $2y_1 + 6y_2 = 0$

⇒ $2y_1 = -6y_2 \Rightarrow y_1 = -3y_2$

$$A^T = \begin{pmatrix} 2 & 6 \\ 5 & 15 \end{pmatrix} \quad R_2 \rightarrow R_2 - 2.5R_1$$

$$\begin{pmatrix} 2 & 6 \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned} y_2 &= 1 \\ y_1 &= -3 \end{aligned}$$

$$\underline{R(A)} = \underline{L(A^T)} = \begin{pmatrix} 2 & 6 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\underline{R(A)} = \text{span} \left\{ \begin{pmatrix} 2 \\ 5 \end{pmatrix} \right\}$$

1

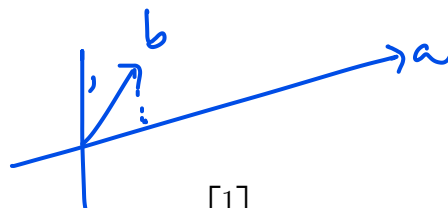
$$A = \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$$

$$\sim \begin{bmatrix} \boxed{1} & 0 & 0 \\ 0 & \boxed{2} & 0 \\ 0 & 0 & \boxed{3} \end{bmatrix}$$

$$\text{Rank} = 3$$

9.

$$P = \frac{aa^T}{a^T a}$$



Which of the following options is/are true when the vector $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ is projected onto the vector $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$?

$b =$

$a =$

A. The projection matrix is $\frac{1}{2} \begin{bmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

✓✓

B. The projection matrix is $\frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$a^T a = [1 \ -1 \ 0] \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= 2$$

✓✓ C. Projected vector is $\frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ ✓✓

D. Projected vector is $\frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ (x)

$$aa^T = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} [1 \ -1 \ 0]_{1 \times 3}$$

3×1

Answer : b, c

$$P = \frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

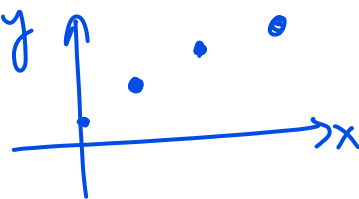
projected vector = $P \cdot b$

$$= \frac{1}{2} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}_{3 \times 1}$$

3×3

$$= \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$



10. Find the best-fit line for the dataset given below using the **least squares method**.

$$y = mx + c$$

x	y
0	1
2	2
3	4
5	3

$$A^T A \hat{x} = A^T b$$

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \\ 3 & 1 \\ 5 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 3 \end{bmatrix}$$

✓ A. $\hat{y} = 1.34 + 0.46x$

B. $\hat{y} = 0.46 + 1.34x$

C. $\hat{y} = 1.34 - 0.46x$

D. $\hat{y} = 2.16 + 0.74x$

Answer : a

$$A^T A = \begin{bmatrix} 0 & 2 & 3 & 5 \\ 1 & 1 & 1 & 1 \end{bmatrix}_{2 \times 4} \begin{bmatrix} 0 & 1 \\ 2 & 1 \\ 3 & 1 \\ 5 & 1 \end{bmatrix}_{4 \times 2}$$

$$= \begin{bmatrix} 38 & 10 \\ 10 & 4 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 0 & 2 & 3 & 5 \\ 1 & 1 & 1 & 1 \end{bmatrix}_{2 \times 4} \begin{bmatrix} 1 \\ 2 \\ 4 \\ 3 \end{bmatrix}_{4 \times 1} = \begin{bmatrix} 31 \\ 10 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 38 & 10 \\ 10 & 4 \end{bmatrix} \hat{x} = \begin{bmatrix} 31 \\ 10 \end{bmatrix} \Rightarrow \begin{bmatrix} 38 & 10 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 31 \\ 10 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} 38 & 10 \\ 10 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 31 \\ 10 \end{bmatrix} = \begin{bmatrix} 1.34 \\ 0.46 \end{bmatrix} \rightarrow \begin{matrix} \text{intercept} \\ \text{slope} \end{matrix}$$

11. Find the best second-order polynomial that fits the following dataset using the least squares method:

$$\min_{\theta_0, \theta_1, \theta_2} \sum_{i=1}^3 \left(\underline{y_i} - \underline{(\theta_0 + \theta_1 x_i + \theta_2 x_i^2)} \right)^2$$

Data:

x	y
1	-1
0	1
-1	1

A. $y = 2 - \frac{1}{2}x + \frac{5}{2}x^2$

B. $y = 2 - \frac{1}{2}x - \frac{5}{2}x^2$

C. $y = 2 + \frac{1}{2}x - \frac{5}{2}x^2$

D. $y = 1 - x - x^2$

$$y = \theta_0 + \theta_1 x + \theta_2 x^2$$

For each data point:

$$y_1 = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2$$

$$y_2 = \theta_0 + \theta_1 x_2 + \theta_2 x_2^2$$

$$y_3 = \theta_0 + \theta_1 x_3 + \theta_2 x_3^2$$

x	y
1	-1
0	1
-1	1

$$\Rightarrow \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 1 & 0 & 0 & 1 \\ 1 & -1 & 1 & 1 \end{array} \right]$$

$$\begin{array}{l} R_1 \rightarrow R_1 - R_2 \\ R_3 \rightarrow R_3 - R_2 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 0 & 1 & 1 & -2 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{array} \right]$$

$$R_1 \leftrightarrow R_2 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & -1 & 1 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2 \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 2 & -2 \end{array} \right]$$

$$2\theta_2 = -2 \Rightarrow \theta_2 = -1$$

$$\theta_1 + \theta_2 = -2 \Rightarrow \theta_1 - 1 = -2 \Rightarrow \theta_1 = -1$$

$$\theta_0 = 1$$

$$y = \theta_0 + \theta_1 x + \theta_2 x^2$$

$$y = 1 - x - x^2$$

x	y
1	-1
0	1
-1	-1